

Task on CloudTrail & CloudWatch

1.Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

Objective

To enable AWS CloudTrail monitoring and store CloudTrail events in both Amazon S3 and Amazon CloudWatch Logs for monitoring and auditing AWS activities.

Configuration Performed

Created a CloudTrail trail named devops-cloudtrail.

Enabled multi-region trail to capture AWS API activity across regions.

Configured an Amazon S3 bucket for storing CloudTrail log files:

S3 Bucket: aws-cloudtrail-logs-634176595009-71a50694

Enabled log file validation for CloudTrail logs.

Created a CloudWatch Log Group:

Log Group: /aws/cloudtrail/devops

Created an IAM role with permission for CloudTrail to write events to CloudWatch Logs.

Configured the CloudTrail trail to send events to the CloudWatch Log Group.

Verified that CloudTrail logging is active and that log files are being delivered to S3.

Result

CloudTrail monitoring was successfully configured. CloudTrail events are stored in Amazon S3 and delivered to Amazon CloudWatch Logs, enabling centralized monitoring and auditing of AWS activities.

us-east-1.console.aws.amazon.com/cloudtrailv2/home?region=us-east-1#/trails/arn:aws:cloudtrail:us-east-1:634176595009:trail/devops-cloudtrail/edit/general...

aws

Search

[Alt+S]

1

United States (N. Virginia)

CHVK (634176595009)

CHVK

CloudTrail > Trails > arn:aws:cloudtrail:us-east-1:634176595009:trail/devops-cloudtrail > Edit

Edit arn:aws:cloudtrail:us-east-1:634176595009:trail/devops-cloudtrail

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

devops-cloudtrail

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location | [Info](#)

☐ Create new S3 bucket
Create a bucket to store logs for the trail.

☒ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket name
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

aws-cloudtrail-logs-634176595009-71a50e94

Browse

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2/log-groups/log-group/\$252Faws\$252Fcloudtrail\$252Fdevops

aws

Search

[Alt+S]

1

United States (N. Virginia)

CHVK (634176595009)

CHVK

CloudWatch > Log management > /aws/cloudtrail/devops

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 0000

▶ AI Operations

▶ GenAI Observability

▶ Application Signals (APM)

▶ Infrastructure Monitoring

▼ Logs

Log Management [New](#)

Log group "/aws/cloudtrail/devops" has been created.

/aws/cloudtrail/devops

ActionsView in Logs InsightsStart tailingSearch log group

▼ Log group details

Log class | [Info](#)
Standard

ARN
arn:aws:logs:us-east-1:634176595009:log-group:/aws/cloudtrail/devops:*

Creation time
35 minutes ago

Retention
Never expire

Stored bytes

Subscription filters
0

Contributor Insights rules
-

KMS key ID
-

Deletion protection
☐ Off

Data protection
-

Custom field indexes
[Configure](#)

Transformer
[Configure](#)

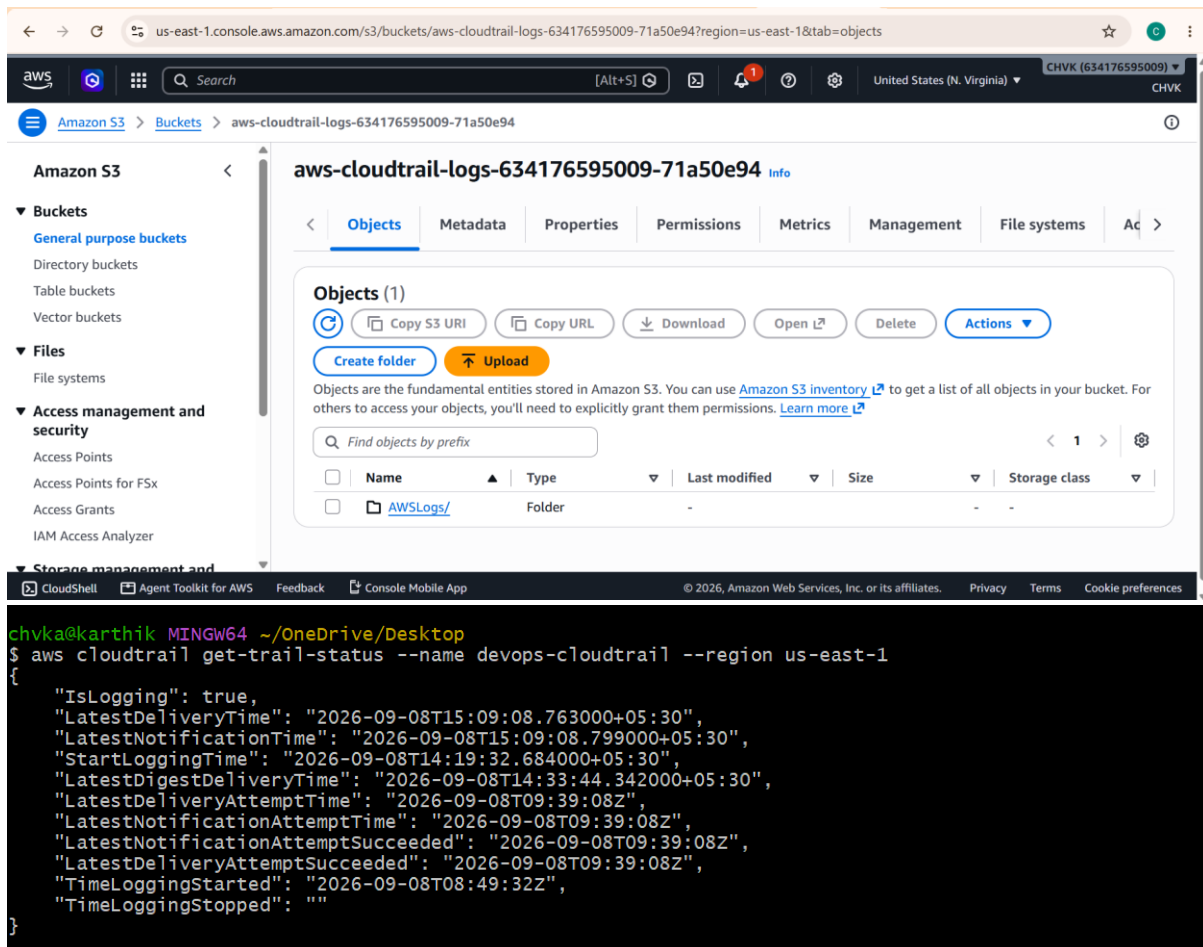
Anomaly detection
[Configure](#)

Bearer token authentication
☐ Off

Syslog ingestion | [Info](#)
[Configure](#)

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences



2.Enable SNS for CloudTrail to send alerts via email.

Objective

To configure Amazon SNS with AWS CloudTrail and receive CloudTrail notifications through email.

Configuration

SNS Topic: aws-cloudtrail-logs-634176595009-3063908c

Protocol: Email

Email: chvkarthik07@gmail.com

Subscription Status: Confirmed

Working

CloudTrail → SNS Topic → Email Notification

CloudTrail sends notifications to the SNS topic, and SNS delivers the notification to the confirmed email address.

Result

The SNS email subscription was successfully created and confirmed. CloudTrail notifications can now be delivered to the confirmed email address.

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/subscription/arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-306...

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

CHVK (634176595009)

Amazon SNS > Topics > aws-cloudtrail-logs-634176595009-3063908c > Subscription: 36327a8e-6b23-4c4a-8418-facef199aaad

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

Subscription to aws-cloudtrail-logs-634176595009-3063908c created successfully.

The ARN of the subscription is arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c:36327a8e-6b23-4c4a-8418-facef199aaad.

Notifications 0 0 1 1 0 0

Subscription: 36327a8e-6b23-4c4a-8418-facef199aaad

Edit Delete

Details

ARN

arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c:36327a8e-6b23-4c4a-8418-facef199aaad

Endpoint

chvkarthik07@gmail.com

Topic

aws-cloudtrail-logs-634176595009-3063908c

Status

Pending confirmation

Protocol

EMAIL

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

sns.us-east-1.amazonaws.com/confirmation.html?TopicArn=arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c&Token=233641...

aws

Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:
arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c:36327a8e-6b23-4c4a-8418-facef199aaad

If it was not your intention to subscribe, [click here to unsubscribe](#).

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/topic/arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c

aws

Search

[Alt+S]

United States (N. Virginia)

CHVK (634176595009)

Amazon SNS > Topics > aws-cloudtrail-logs-634176595009-3063908c

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

Name

aws-cloudtrail-logs-634176595009-3063908c

ARN

arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c

Display name

-

Type

Standard

Topic owner

634176595009

Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Subscriptions (2)

Edit

Delete

Request confirmation

Confirm subscription

Create subscription

Search

< 1 >

ID	Endpoint	Status	Protocol
36327a8e-6b23-4c4a-8...	chvkarthik07@gmail.com	Confirmed	EMAIL

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

Objective

To configure Amazon CloudWatch to monitor EC2 instance metrics such as CPU utilization.

Configuration

EC2 Instance: Private-Tomcat-Server

Instance ID: i-00b0acbd2616c3812

Region: US East (N. Virginia)

Metric: CPUUtilization

Dashboard Name: EC2-Monitoring

Implementation

Started the Private-Tomcat-Server EC2 instance.

Opened Amazon CloudWatch.

Navigated to CloudWatch Metrics → Classic Metrics.

Selected EC2 Instance Metrics.

Selected the CPUUtilization metric for the Private-Tomcat-Server instance.

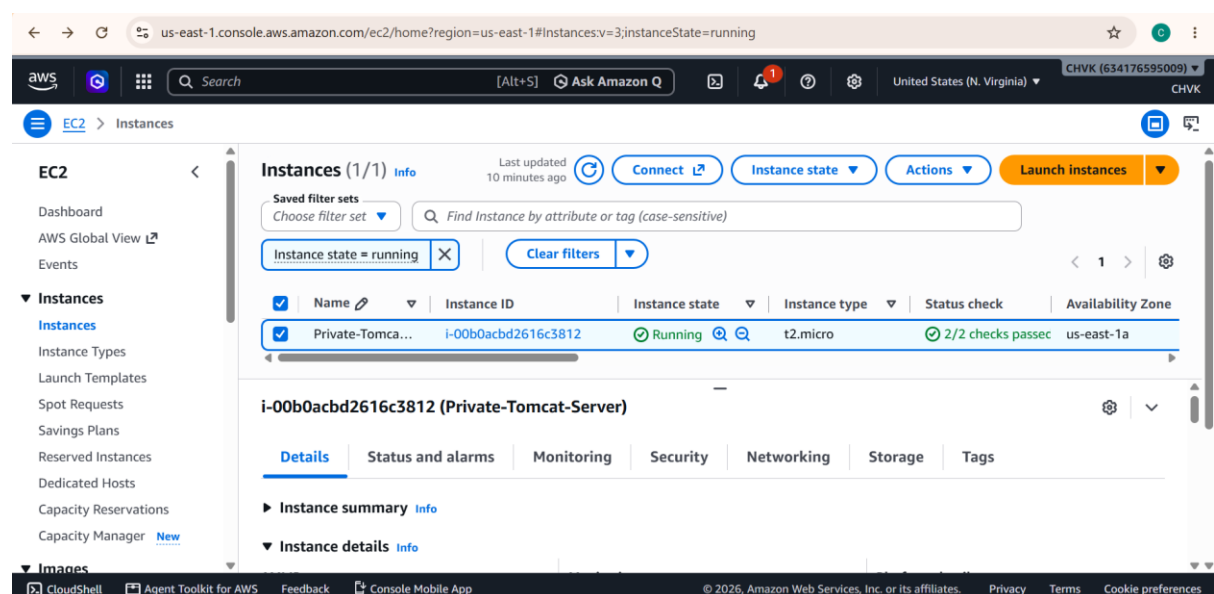
Created a CloudWatch dashboard named EC2-Monitoring.

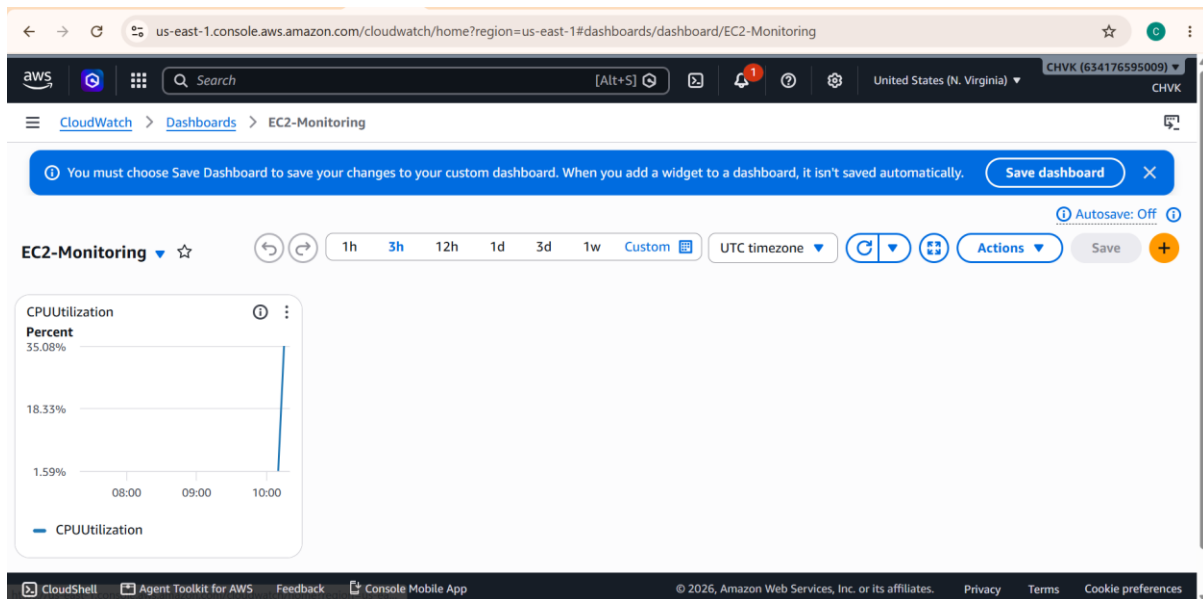
Added the CPUUtilization metric to the dashboard.

Saved the dashboard.

Result

The EC2 instance CPU utilization is successfully monitored using Amazon CloudWatch, and the CPUUtilization metric has been added to the EC2-Monitoring dashboard.





4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Objective

To create a CloudWatch alarm that sends an email notification when the EC2 instance CPU utilization reaches or exceeds 70%.

Configuration Details

EC2 Instance: Private-Tomcat-Server

Instance ID: i-00b0acbd2616c3812

Metric: CPUUtilization

Threshold: 70%

Condition: Greater than or equal to 70%

Evaluation Period: 1 datapoint within 5 minutes

Alarm Name: EC2-CPU-Above-70

Notification: Amazon SNS

SNS Topic: aws-cloudtrail-logs-634176595009-3063908c

Implementation

Opened Amazon CloudWatch.

Navigated to Alarms → Create alarm.

Selected the CPUUtilization metric for the EC2 instance.

Configured the alarm condition as CPU utilization $\geq$ 70%.

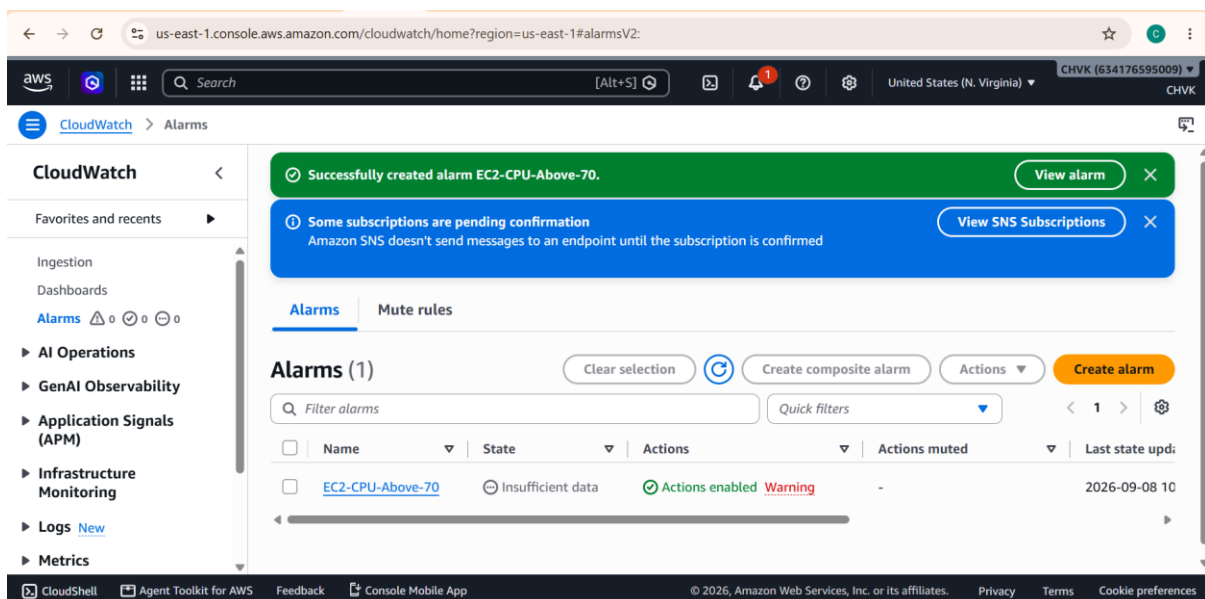
Configured the SNS notification for the In Alarm state.

Provided the alarm name as EC2-CPU-Above-70.

Created the CloudWatch alarm successfully.

Result

The CloudWatch alarm was successfully created. When the CPU utilization of the EC2 instance reaches or exceeds 70%, CloudWatch triggers the alarm and sends a notification through Amazon SNS.



5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

Objective

To monitor the Tomcat service running on an EC2 instance using Amazon CloudWatch and create an alarm to send an SNS email notification when the Tomcat service is not running.

EC2 Instance Details

Instance Name: tomcat-monitor

Instance ID: i-0b6608b154d06a962

Instance Type: t2.micro

Operating System: Amazon Linux 2023

Private IP: 172.31.94.218

Region: US East (N. Virginia)

Tomcat Configuration

Apache Tomcat Version: 9.0.109

Java Version: OpenJDK 17 (Amazon Corretto)

Tomcat Service: tomcat.service

Tomcat Installation Path: /opt/tomcat

Tomcat was configured as a systemd service and verified to be running successfully.

CloudWatch Agent Configuration

The Amazon CloudWatch Agent was installed and configured to monitor the Tomcat process using the procstat plugin.

The monitored process pattern was:

org.apache.catalina.startup.Bootstrap

The custom CloudWatch namespace created was:

TomcatMonitoring

The metric created was:

procstat_lookup_pid_count

This metric represents the number of running Tomcat processes.

Dashboard Monitoring

The Tomcat process metric was added to CloudWatch monitoring.

Namespace: TomcatMonitoring

Metric: procstat_lookup_pid_count

Process Pattern: org.apache.catalina.startup.Bootstrap

When the metric value is 1 or greater, the Tomcat process is running.

When the metric value becomes 0, the Tomcat process is not running.

CloudWatch Alarm

An alarm was created to detect when the Tomcat service is down.

Alarm Name: tomcat-service-down

Metric: procstat_lookup_pid_count

Statistic: Average

Period: 5 minutes

Condition: Lower than 1

Alarm State Trigger: In alarm

Notification: Amazon SNS

The alarm triggers when the Tomcat process count falls below 1, indicating that the Tomcat service is not running.

SNS Notification

The existing Amazon SNS topic was configured to send email notifications when the CloudWatch alarm enters the INSUFFICIENT DATA/ALARM state as applicable.

SNS Topic:

aws-cloudtrail-logs-634176595009-3063908c

Result

Tomcat service monitoring was successfully configured using Amazon CloudWatch Agent and the procstat metric. A CloudWatch alarm named tomcat-service-down was successfully created to monitor Tomcat availability and send notifications through Amazon SNS when the Tomcat process is not running.

```
● tomcat.service - Apache Tomcat
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 11:33:22 UTC; 19s ago
   Process: 27275 /etc/startup/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 27282 (java)
   Tasks: 30 (limit: 1113)
   Memory: 69.3M
   CPU: 2.703s
   CGroup: /system.slice/tomcat.service
           └─27282 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apac

Sep 08 11:33:22 ip-172-31-94-218.ec2.internal systemd[1]: Starting tomcat.service - Apache Tomcat...
Sep 08 11:33:22 ip-172-31-94-218.ec2.internal startup.sh[27275]: Tomcat started.
Sep 08 11:33:22 ip-172-31-94-218.ec2.internal systemd[1]: Started tomcat.service - Apache Tomcat.
```

```
[ec2-user@ip-172-31-94-218 opt]$ sudo dnf install amazon-cloudwatch-agent -y
Last metadata expiration check: 0:10:49 ago on Tue Sep 8 11:23:37 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
amazon-cloudwatch-agent	x86_64	1.300069.1-1.amzn2023	amazonlinux	69 M

Transaction Summary

Install 1 Package

Total download size: 69 M

Installed size: 235 M

Downloading Packages:

amazon-cloudwatch-agent-1.300069.1-1.amzn2023.x86_64.rpm	63 MB/s 69 MB	00:01
--	-----------------	-------

Total

	61 MB/s 69 MB	00:01
--	-----------------	-------

Running transaction check

Transaction check succeeded.

Running transaction test

Transaction test succeeded.

Running transaction

Preparing	:	1/1
Running scriptlet: amazon-cloudwatch-agent-1.300069.1-1.amzn2023.x86_64	:	1/1
create group cwagent, result: 0	:	
create user cwagent, result: 0	:	
Installing	: amazon-cloudwatch-agent-1.300069.1-1.amzn2023.x86_64	1/1
Running scriptlet: amazon-cloudwatch-agent-1.300069.1-1.amzn2023.x86_64	:	1/1
Verifying	: amazon-cloudwatch-agent-1.300069.1-1.amzn2023.x86_64	1/1

```
● amazon-cloudwatch-agent.service - Amazon Cloudwatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 11:38:28 UTC; 27s ago
   Main PID: 27766 (/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml -envconfig /opt/
   Tasks: 6 (limit: 1113)
   Memory: 46.3M
   CPU: 241ms
   CGroup: /system.slice/amazon-cloudwatch-agent.service
           └─27766 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml -envconfig /opt/

Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27770]: Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments:
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: DI [EC2] Found active network interface
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: I! Imds retry client will retry 1 times! Detected the instance is EC2
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: 2026/09/08 11:38:28 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json does not exist or canb
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: 2026/09/08 11:38:28 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: 2026/09/08 11:38:28 I! Valid json input schema.
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: I! Trying to detect region from ec2
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27774]: 2026/09/08 11:38:28 Configuration validation first phase succeeded
Sep 08 11:38:28 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[27766]: I! Detecting run_as_user...

lines 1-20/20 (END)
[ec2-user@ip-172-31-94-218 opt]$ sudo systemctl restart amazon-cloudwatch-agent
[ec2-user@ip-172-31-94-218 opt]$ client_loop: send disconnect: Connection reset by peer
```

```
[ec2-user@ip-172-31-94-218 ~]$ sudo grep -i "procstat[|error|fail]" /opt/aws/amazon-cloudwatch-agent/logs/amazon-cloudwatch-agent.log | tail -30
[[inputs.procstat]]
  telegraf_procstat/841651561:
    - telegraf_procstat/841651561
2026-09-08T11:40:24Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:40:24Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
[[inputs.procstat]]
  telegraf_procstat/841651561:
    - telegraf_procstat/841651561
2026-09-08T11:43:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:43:12Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:44:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:44:13Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:45:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:45:13Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:46:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:46:14Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:47:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:47:15Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:48:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:48:17Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:49:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:49:44Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
2026-09-08T11:50:12Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
[[inputs.procstat]]
  telegraf_procstat/841651561:
    - telegraf_procstat/841651561
2026-09-08T11:52:47Z E! cloudwatch: code: AccessDenied, message: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not authorized to perform
cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action, original error: <nil>
2026-09-08T11:52:47Z E! cloudwatch: writeToCloudwatch failure, err: AccessDenied: User: arn:aws:sts:634176595009:assumed-role/EC2-SSM-role/i-0b6608b154d06a962 is not auth
orized to perform: cloudwatch:PutMetricData because no identity-based policy allows the cloudwatch:PutMetricData action
```

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:

CloudWatch > Alarms

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 0 1 1

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs New

Metrics

Successfully created alarm tomcat-service-down. View alarm

Some subscriptions are pending confirmation
Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed. View SNS Subscriptions

Alarms Mute rules

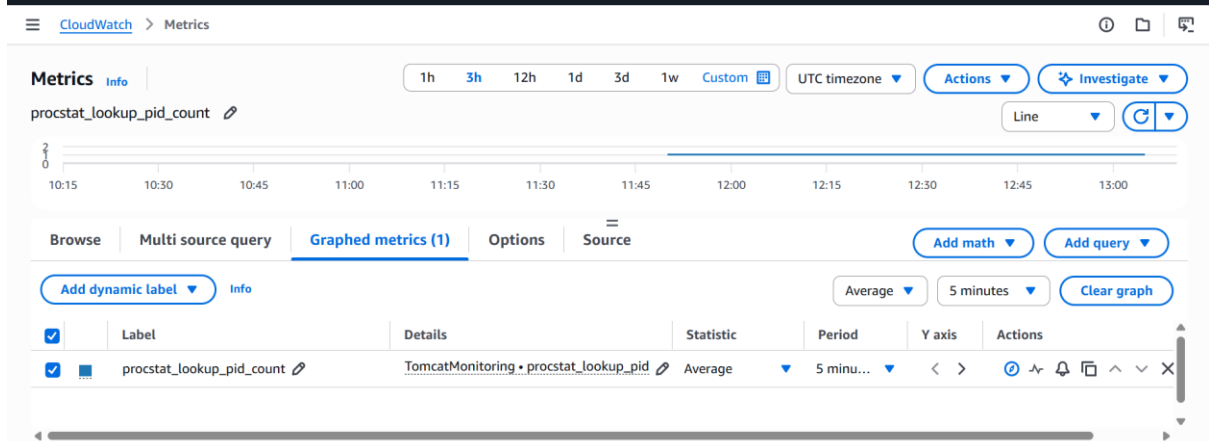
Alarms (2)

Clear selection Create composite alarm Actions Create alarm

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted	Last state update
☐	tomcat-service-down	OK	Actions enabled Warning	-	2026-09-08 13
☐	EC2-CPU-Above-70	Insufficient data	Actions enabled Warning	-	2026-09-08 11

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#topic/arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c

Amazon SNS > Topics > aws-cloudtrail-logs-634176595009-3063908c

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

aws-cloudtrail-logs-634176595009-3063908c Edit Delete Publish message

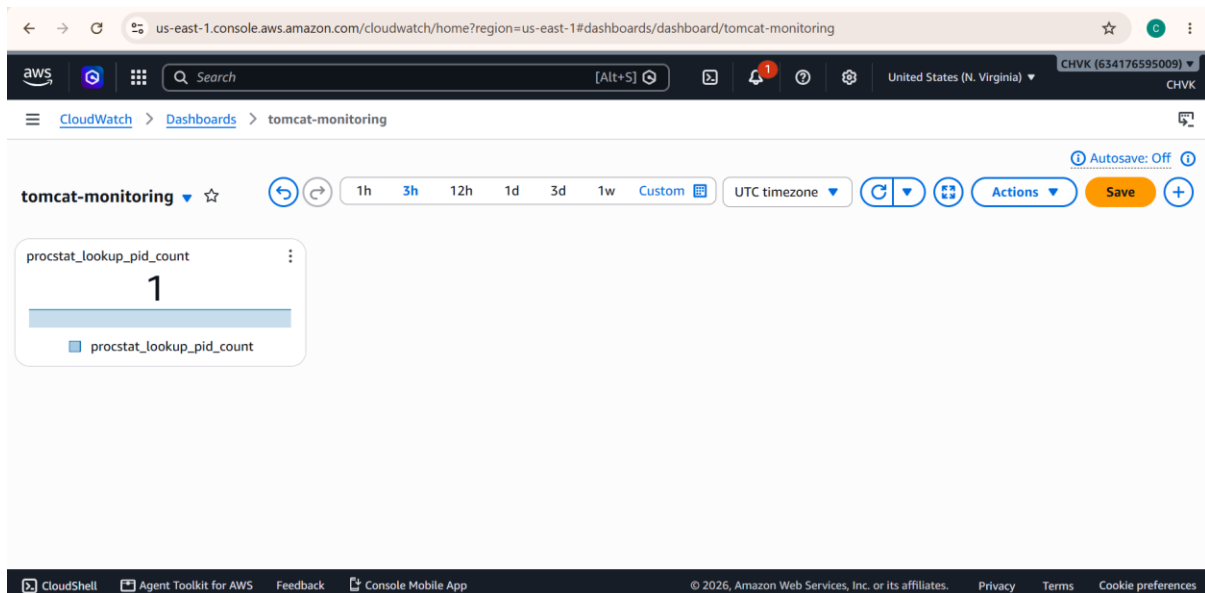
Details

Name aws-cloudtrail-logs-634176595009-3063908c	ARN arn:aws:sns:us-east-1:634176595009:aws-cloudtrail-logs-634176595009-3063908c	Display name -	Type Standard
Topic owner 634176595009			

< Subscriptions Access policy Delivery policy (HTTP/S) Delivery status logging Encryption >

Subscriptions (2) Edit Delete Request confirmation Confirm subscription Create subscription

Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Objective

To create a CloudWatch Dashboard to monitor the Nginx service and configure an alarm to send an alert when the Nginx service is not running.

EC2 Instance Details

Instance Name: tomcat-monitor

Instance ID: i-0b6608b154d06a962

Instance Type: t2.micro

Region: US East (N. Virginia)

Private IP: 172.31.94.218

Public IP: 54.89.247.129

Nginx Configuration

Nginx was installed on the EC2 instance and configured as a system service.

Commands used:

```
sudo dnf install nginx -y
```

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```

```
sudo systemctl status nginx
```

The Nginx service was successfully started and enabled.

CloudWatch Agent Configuration

The Amazon CloudWatch Agent was configured to monitor the Nginx process using procstat.

CloudWatch Namespace: NginxMonitoring

Process Pattern: nginx

Metric: procstat_lookup_pid_count

The metric is used to identify whether the Nginx process is running.

CloudWatch Dashboard

Dashboard Name: nginx-monitoring

The Nginx process metric was added to the CloudWatch Dashboard to monitor the service continuously.

CloudWatch Alarm

Alarm Name: nginx-alarm

Alarm Configuration:

Metric: procstat_lookup_pid_count

Statistic: Average

Period: 5 minutes

Condition: Lower than 1

Evaluation: 1 datapoint out of 1

Alarm State: In Alarm when the Nginx process count is below 1

Notification: Amazon SNS

SNS Topic: aws-cloudtrail-logs-634176595009-3063908c

Result

Nginx service monitoring was successfully configured using Amazon CloudWatch. A CloudWatch Dashboard was created to monitor the Nginx process, and an alarm was configured to send an SNS notification when the Nginx service is not running.

```
[ec2-user@ip-172-31-94-218 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 13:19:35 UTC; 27s ago
     Main PID: 31742 (nginx)
       Tasks: 2 (limit: 1113)
      Memory: 2.8M
         CPU: 38ms
    CGroup: /system.slice/nginx.service
            └─31742 "nginx: master process /usr/sbin/nginx"
              └─31743 "nginx: worker process"

Sep 08 13:19:35 ip-172-31-94-218.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 08 13:19:35 ip-172-31-94-218.ec2.internal nginx[31740]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 08 13:19:35 ip-172-31-94-218.ec2.internal nginx[31740]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 08 13:19:35 ip-172-31-94-218.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-94-218 ~]$

● amazon-cloudwatch-agent.service - Amazon Cloudwatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 13:23:39 UTC; 1min 24s ago
     Main PID: 32206 (amazon-cloudwat)
       Tasks: 7 (limit: 1113)
      Memory: 52.1M
         CPU: 374ms
    CGroup: /system.slice/amazon-cloudwatch-agent.service
            └─32206 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml -envconfig /opt

Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32210]: Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments:
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: D! [Ec2] Found active network interface
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: I! Imds retry client will retry 1 times! Detected the instance is EC2
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: 2026/09/08 13:23:39 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json does not exist or can
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: 2026/09/08 13:23:39 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: 2026/09/08 13:23:39 I! Valid json input schema.
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: I! Trying to detect region from ec2
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32214]: 2026/09/08 13:23:39 Configuration validation first phase succeeded
Sep 08 13:23:39 ip-172-31-94-218.ec2.internal start-amazon-cloudwatch-agent[32206]: I! Detecting run_as_user...
lines 1-20/20 (END)
```

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:

CloudWatch > Alarms

Successfully created alarm nginx-alarm. View alarm

Some subscriptions are pending confirmation
Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed. View SNS Subscriptions

Alarms Mute rules

Alarms (3)

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted	Last state update
☐	nginx-alarm	Insufficient data	Actions enabled Warning	-	2026-09-08 13
☐	tomcat-service-down	OK	Actions enabled Warning	-	2026-09-08 13

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#metricsV2?graph=~(view~timeSeries~stacked~false~metrics~{~(TomcatMonitorin...

CloudWatch > Metrics

Metrics Info

1h 3h 12h 1d 3d 1w Custom UTC timezone Actions Investigate

procstat_lookup_pid_count

Line

Browse (1) Multi source query Graphed metrics (1) Options Source

Alarm recommendations Download alarm code Create alarm Graph with SQL Graph search

N. Virginia

Search for any metric, dimension, resource id or account id

☐	host 1/1	pattern	pid_finder	Metric name	Alarms
☐	ip-172-31-94-218.ec2.internal	nginx	native	procstat_lookup_pid_count	No alarms

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards/dashboard/nginx-monitoring

CloudWatch > Dashboards > nginx-monitoring

Autosave: Off

nginx-monitoring

1h 3h 12h 1d 3d 1w Custom UTC timezone Actions Save

procstat_lookup_pid_count

2

procstat_lookup_pid_count